THE STRICT LAYER OF DE MORGAN CUBICAL SETS: MINIMAL FILLERS, A COHERENCE DICHOTOMY, AND THE KAROUBI ENVELOPE

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ABSTRACT. The De Morgan cube category $\square_{\mathrm{DM}}$ — cubes with faces, degeneracies, symmetries, both connections, reversal, and diagonals, the shape category of CCHM cubical type theory — is not an Eilenberg–Zilber category: we show it is not even idempotent complete, and identify its Karoubi envelope as the opposite of the category of finite *projective* De Morgan algebras, which a theorem of Bova and Cabrer characterizes combinatorially — a bridge between the cubical and the unification-theoretic literatures that appears not to have been noticed. We then analyze the structure that survives the failure of Eilenberg–Zilber normal forms. Building on a companion study of boundary fibers in free De Morgan algebras (fibers are Boolean intervals 2^D over antichains of diagonal points), we prove a one-step *defect formula*: deleting the minimal diagonal points of any element reaches the least element of its boundary fiber, so the failure of any substitution to preserve minimality is measured by an explicit antichain. Our main result is a *coherence dichotomy*: for a substitution f and an element g, correcting before or after transport agree if and only if the transported boundary avoids the defect antichain of g — an equational criterion that holds unconditionally on the wide subcategory generated by faces, symmetries and reversal, and fails in a controlled way for degeneracies, connections and diagonals; in particular the minimal skeleton is not even cylinder-stable. All statements are verified exhaustively by machine through dimension three (7,828,354 elements). We conclude that in the absence of Eilenberg–Zilber structure, thinness of fillers cannot be a property and must be carried as structure — the median interpolation of the companion paper being the minimal such structure.

1. Introduction

Simplicial sets owe much of their combinatorial control to the Eilenberg–Zilber lemma: every cell is uniquely a degeneracy of a nondegenerate cell. Cube categories with connections but without diagonals enjoy the same property [3]; the shape category of Cohen–Coquand–Huber–Mörtberg cubical type theory [5] — cubes governed by the free De Morgan algebra, with diagonals and reversal — does not. This paper is about what replaces it.

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Write $\square_{\mathrm{DM}}$ for the De Morgan cube category: objects $[n]$, maps $[m] \to [n]$ the n-tuples of elements of the free De Morgan algebra DM_m; equivalently, the opposite of the category of finitely generated free De Morgan algebras. Presheaves on $\square_{\mathrm{DM}}$ are the De Morgan cubical sets interpreting CCHM type theory, and the maps of $\square_{\mathrm{DM}}$ are exactly the *reparametrizations* that a proof assistant's kernel applies silently: the *strict layer* of the theory, as opposed to the weak layer of Kan compositions. A companion paper [10] determined the boundary fibers of DM_n: the set of elements with prescribed faces is a Boolean interval 2^D, where D is an antichain of *diagonal* points of the literal cube, with extremal fibers of size $2^{\binom{n}{\lfloor n/2 \rfloor}}$ by Sperner's theorem. Thus a boundary does not determine its filler — degenerate-style cells are non-unique in a quantified, Boolean-graded way. The present paper turns that local computation into a global structure theory for $\square_{\mathrm{DM}}$, in three steps.
The Karoubi envelope (§3). First, the negative fact in its sharpest form: $\square_{\mathrm{DM}}$ is not idempotent complete — the idempotent $x \mapsto x \vee \neg x$ on [1] does not split (Proposition 3.1), so $\square_{\mathrm{DM}}$ is not a generalized Reedy category and in particular not Eilenberg–Zilber, by the same first obstruction Campion [3] exhibited for the diagonal–connection sites. Campion's classification, however, never meets $\square_{\mathrm{DM}}$: his cube categories are categories of $\{0,1\}$-valued operations, so his full-signature site is the *Boolean* cube category, where $x \vee \neg x = 1$. We identify the Karoubi envelope of the genuinely De Morgan site:

$$\mathrm{Kar}(\square_{\mathrm{DM}}) \;\simeq\; (\text{finite projective De Morgan algebras})^{\mathrm{op}},$$

and finite projective De Morgan algebras were characterized combinatorially by Bova and Cabrer [1] in the theory of unification: dually, involutive posets satisfying three explicit conditions. The two literatures do not cite each other; the identification, though easy once stated, appears to be new, and slots $\square_{\mathrm{DM}}$ into the table of cubical Karoubi envelopes next to Sattler's Dedekind computation [9] and Campion's Boolean one.
The defect formula (§4). Second, the positive replacement for degeneracy normal forms. Every $g \in \mathrm{DM}_n$ determines the antichain A_g of its minimal *diagonal* points, and we prove (Theorem 4.1) that

$$\mathrm{r}(g) \;:=\; g \setminus A_g$$

is, in a single step, the least element of the boundary fiber of g. Consequently, for any substitution f and any boundary-minimal φ, the failure of $f^*\varphi$ to remain minimal is measured by the explicit antichain $A_{f^*\varphi}$ — the *defect* — and corrected by deleting it, a move that stays inside one Boolean fiber and is therefore connected to the uncorrected transport by the strict median cylinder of [10].
The coherence dichotomy (§5). Third, the main theorem. Correcting commutes with transporting — $\mathrm{r}(f^*g) = \mathrm{r}(f^*\mathrm{r}(g))$ — if and only if the transported boundary never reads a defect point:

Theorem (Theorem 5.1)**.** For a substitution $f : [m] \to [n]$ with induced map F of literal cubes, and $g \in \mathrm{DM}_n$, the following are equivalent:

(1) $\mathrm{r}(f^*g) = \mathrm{r}(f^*\mathrm{r}(g))$;
(2) f^*g and $f^*\mathrm{r}(g)$ have equal boundaries;
(3) $F(\mathrm{Face}(Q_m)) \cap A_g = \emptyset$.

Each implication has a two-line proof, and the trichotomy of generators falls out: faces, symmetries, reversal (and the square-degeneracies $x \mapsto x \wedge \neg x$) send face points to face points, so on the wide subcategory they generate the minimal skeleton is a strictly natural retraction; degeneracies, connections and diagonals map some face points onto diagonals, and for them the criterion is sharp. Two consequences deserve emphasis. The exceptional set is genuinely nonempty already for *degeneracies*: the cylinder $g \times I$ has g itself among its faces, so the minimal skeleton is **not cylinder-stable** — a precise expression of how far the strict layer is from a Reedy-style theory. And the criterion is exact, not merely sufficient: on all 7,828,354 elements of DM_3 the three conditions of the theorem carve out the same 1,133,756 failures for a diagonal and for a connection substitution (§7).
Thinness as structure (§6). In cubical ω-categories with connections, thin fillers of commutative shells exist *uniquely* (Higgins [8], Steiner [11]); the uniqueness depends on strict composition and on the normal forms of an Eilenberg–Zilber site, and Steiner anticipated that in weak settings thin fillers would exist "but not necessarily" uniquely. The De Morgan strict layer realizes that anticipation in a free, composition-less setting: fillers form Boolean torsors, no property singles one out functorially (the dichotomy), and the canonical data that remains — minimal filler, defect antichain, median cylinder — is *structure*. We take this as the conceptual summary: *where Eilenberg–Zilber fails, thinness cannot be a property; the median-term algebra is the minimal structure replacing it.*
Relation to other work. Buchholtz–Morehouse [2] proved that the De Morgan cube category is a *strict test category*, so the homotopy theory over $\square_{\mathrm{DM}}$ is standard — our subject is the strict layer that presents it, which their analysis leaves untouched. Cavallo–Sattler [4] eliminate reversals up to homotopy over self-dual interval theories; their twist construction $T(I) = I \times I$ is, on representables, exactly the literal-cube presentation used here (the diagonal points being the twist diagonal), and their results are complementary: elimination is up-to-path and model-level, while the judgmental layer of implemented proof assistants remains De Morgan. Doré–Cavallo–Mörtberg [7] search for single fillers over the same algebras; the present results describe the whole solution set and its (non-)functoriality. All machine verification was performed with the scripts accompanying [10], extended for this paper (§7).

2. Preliminaries

2.1. **The De Morgan cube category.** DM_n denotes the free De Morgan algebra on $x_1, \dots, x_n$; we freely use its classical presentation (see [10] and the references there): the lattice of monotone Boolean functions on the *literal cube* $Q_n = \{0,1\}^{2n}$, coordinates grouped in pairs $(x_i, \neg x_i)$, elements identified with up-sets. A point is a *face point* if some pair equals $(0,1)$ or $(1,0)$, a *diagonal point* if every pair is $(0,0)$ or $(1,1)$; write $\mathrm{Face}(Q_n)$ and $\mathrm{Diag}(Q_n)$. The *faces* $\partial_i^e : \mathrm{DM}_n \to \mathrm{DM}_{n-1}$ substitute $x_i := e$; the *boundary* ∂g is the tuple of all faces.

The *De Morgan cube category* $\square_{\mathrm{DM}}$ has objects $[n]$ $(n \geq 0)$ and morphisms $[m] \to [n]$ the n-tuples $f = (t_1, \dots, t_n)$, $t_i \in \mathrm{DM}_m$, composing by substitution; it is the opposite of the category of finitely generated free De Morgan algebras, and contains faces, degeneracies, symmetries, both connections ($x_i := x_i \wedge x_j$, $x_i := x_i \vee x_j$), reversal ($x_i := \neg x_i$), and diagonals ($x_i := x_j$). A morphism f induces the substitution $f^* : \mathrm{DM}_n \to \mathrm{DM}_m$ and the *literal-cube map* $F : Q_m \to Q_n$, $F(\alpha)_i = \big(t_i(\alpha), (\neg t_i)(\alpha)\big)$, so that $(f^*g)(\alpha) = g(F(\alpha))$. Note that F is a map of *all* points and need not preserve the face/diagonal dichotomy; this freedom is where everything below happens.

2.2. **The fiber theorem.** We import from [10]: every fiber of the boundary map on DM_n is an interval $[m, M]$ whose difference $D = M \setminus m$ is an antichain of diagonal points; the fiber is Boolean, canonically 2^D; every antichain of diagonal points arises; each compatible sphere has a least filler (the up-closure of its prescribed points); boundaries are in bijection with antichains of face points; and for same-boundary φ, φ' the median interpolation $\chi = (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$ is a cylinder in one fresh variable, constant on all faces. An element is *boundary-minimal* if it is the least element of its fiber, equivalently if all its minimal points are face points.

3. The Karoubi envelope

Proposition 3.1. *The endomorphism $e = (x \mapsto x \vee \neg x)$ of $[1]$ is idempotent and does not split. Hence $\square_{\mathrm{DM}}$ is not idempotent complete, not a generalized Reedy category, and not an Eilenberg–Zilber category.*

Proof. Idempotency is the absorption $(x \vee \neg x) \vee \neg(x \vee \neg x) = (x \vee \neg x) \vee (\neg x \wedge x) = x \vee \neg x$. Suppose $e = s \circ p$ with $p : [1] \to [k]$, $s : [k] \to [1]$ and $p \circ s = \mathrm{id}_{[k]}$. Contravariantly $s^* \circ p^* = \mathrm{id}_{\mathrm{DM}_k}$, so DM_k is a retract of DM_1; since $|\mathrm{DM}_1| = 6$ and $|\mathrm{DM}_k| = M(2k) \geq 168$ for $k \geq 2$, we must have $k \leq 1$. For $k = 0$, e factors through $[0]$, so e^* is constant; but $x \vee \neg x \notin \{0,1\}$ in DM_1. For $k = 1$, the retraction forces s^* to be bijective (finite cardinality), so $e^* = p^* \circ s^*$ is injective; but $e^*(x) = x \vee \neg x = e^*(x \vee \neg x)$ while $x \neq x \vee \neg x$. The last two claims follow since generalized Reedy categories are idempotent complete [3]. $\square$

Theorem 3.2. *The Karoubi envelope of the De Morgan cube category is the opposite of the category of finite projective De Morgan algebras:*

$$\mathrm{Kar}(\square_{\mathrm{DM}}) \;\simeq\; (\mathsf{DM}\text{-}\mathsf{proj}_{\mathrm{fin}})^{\mathrm{op}}.$$

Its objects are characterized combinatorially by Bova–Cabrer [1, Thm. 11]*: under Cornish–Fowler duality* [6]*, a finite De Morgan algebra is projective iff its dual involutive poset* $(P, \leq, i)$ *is a nonempty lattice, every* $x \leq i(x)$ *admits* $y \geq x$ *with* $y = i(y)$*, and the sub-poset* $\{x \mid x \leq i(x)\}$ *is* 3*-complete.*

Proof. $\square_{\mathrm{DM}}$ is the opposite of the finitely generated free De Morgan algebras. In any variety, idempotents split in the category of algebras, and retracts of free algebras are exactly the projectives; hence the idempotent completion of the full subcategory of f.g. frees is the full subcategory of f.g. projectives. The variety of De Morgan algebras is locally finite, so finitely generated projectives are finite [1]. Taking opposites gives the displayed equivalence, and the characterization of the objects is Bova–Cabrer's theorem. □

Remark 3.3. This places $\square_{\mathrm{DM}}$ in the table of cubical Karoubi envelopes: Dedekind cubes with diagonals complete to finite lattices with monotone maps (Sattler [9], cf. [3, Prop. 8.8]); Boolean cubes to finite nonempty sets [3, Prop. 8.11]; De Morgan cubes to the dual of finite projective De Morgan algebras. The unification-theoretic literature where the projectivity characterization lives [1] and the cubical literature appear never to have cited one another. The same argument identifies the Karoubi envelope of the *Kleene* cube category via [1, Thm. 12]. We note that Campion's negative results [3, Thm. 8.12(2)] make it unlikely that any degree function turns these envelopes into Eilenberg–Zilber categories; the sequel does not attempt this, and instead isolates what survives.

4. Minimal fillers: the defect formula

For $g \in \mathrm{DM}_n$ let A_g be the set of minimal points of g (qua up-set) that are diagonal. Minimal points form an antichain, so A_g is an antichain of diagonal points.

Theorem 4.1 (defect formula)**.** *For every* $g \in \mathrm{DM}_n$*, the element* $\mathrm{r}(g) := g \setminus A_g$ *is the least element of the boundary fiber of* g*. In particular one deletion step reaches the fiber minimum, and* g *is boundary-minimal iff* $A_g = \emptyset$*.*

Proof. $\mathrm{r}(g)$ is an up-set: a point strictly above $\alpha \in A_g$ is either a face point (hence in g, hence in $\mathrm{r}(g)$), or a diagonal point $\beta > \alpha$, in which case the companion lemma [10, Lem. 2.3] provides a face point γ with $\alpha < \gamma \leq \beta$; then $\gamma \in g$, so $\beta \in {\uparrow}\gamma \subseteq \mathrm{r}(g)$. Its boundary equals ∂g since only diagonal points were removed. No new diagonal minimal points appear: if β is diagonal and minimal in $\mathrm{r}(g)$, then β was not minimal in g (else $\beta \in A_g$), so some $\delta < \beta$ lies in g; taking δ minimal in g, either δ is a face point — but then $\delta \in \mathrm{r}(g)$, contradicting minimality of β — or $\delta \in A_g$ is diagonal, and the companion lemma gives a face point $\gamma \in (\delta, \beta]$, with $\gamma \neq \beta$ (β is diagonal), lying in $\mathrm{r}(g)$ — again a contradiction. So all minimal points of $\mathrm{r}(g)$ are face points, which characterizes the least element of the fiber [10, Thm. 4.2]. $\square$

Corollary 4.2 (the defect of a substitution)**.** *Let $f : [m] \to [n]$ be a morphism of $\square_{\mathrm{DM}}$ and $\varphi \in \mathrm{DM}_n$ boundary-minimal. Then the deviation of $f^*\varphi$ from boundary-minimality is the explicit antichain*

$$A_{f^*\varphi} \;=\; \{\, \alpha \in \mathrm{Diag}(Q_m) \;:\; F(\alpha) \in \varphi \text{ and } F(\beta) \notin \varphi \text{ for every lower cover } \beta \lessdot \alpha \,\},$$

and $\mathrm{r}(f^*\varphi)$ differs from $f^*\varphi$ exactly by this antichain, inside one Boolean fiber; the median cylinder of [10] *connects the two by a strict homotopy constant on the boundary.*

5. The coherence dichotomy

The retraction r is a normal form for *fixed* $[n]$; the question is its interaction with substitution. Call $f : [m] \to [n]$ and $g \in \mathrm{DM}_n$ *coherent* if $\mathrm{r}(f^*g) = \mathrm{r}(f^*\mathrm{r}(g))$.

Theorem 5.1 (coherence dichotomy)**.** *For every morphism f of $\square_{\mathrm{DM}}$ and every $g \in \mathrm{DM}_n$, the following are equivalent:*

1. *(f, g) is coherent;*
2. *$\partial(f^*g) = \partial(f^*\mathrm{r}(g))$;*
3. *$F(\mathrm{Face}(Q_m)) \cap A_g = \emptyset$.*

Proof. (3)⇒(2): g and $\mathrm{r}(g)$ agree outside A_g, and the boundaries of the transports only read the values of g, resp. $\mathrm{r}(g)$, on $F(\mathrm{Face}(Q_m))$. (2)⇒(1): elements with equal boundaries lie in one fiber, whose least element is unique. (1)⇒(3), contrapositively: if α is a face point with $F(\alpha) \in A_g$, then $(f^*g)(\alpha) = g(F(\alpha)) = 1$ while $(f^*\mathrm{r}g)(\alpha) = 0$; a disagreement at a face point is a disagreement of boundaries, and least elements determine their boundaries ($\mathrm{r}(a) = \mathrm{r}(b)$ forces $\partial a = \partial\mathrm{r}(a) = \partial\mathrm{r}(b) = \partial b$), so the two corrected elements differ. $\square$

Proposition 5.2 (classification of generators)**.** *Faces, symmetries, reversal, and the substitutions $x_i := x_i \wedge \neg x_i$, $x_i := x_i \vee \neg x_i$ send face points to face points; hence they are coherent for* every *g, and the wide subcategory $\square^{\mathrm{fp}}_{\mathrm{DM}} \subseteq \square_{\mathrm{DM}}$ of face-point-preserving morphisms (which they generate, and which is closed under composition) carries* r *as a strictly natural retraction onto boundary-minimal elements. Degeneracies, connections, and diagonals each map some face point to a diagonal point, and for each of them elements g violating condition (3) exist.*

Proof. For a face $[n-1] \to [n]$, F embeds Q_{n-1} into a face subcube, sending every point (facial or not) to a face point. Symmetries and reversal permute pair-coordinates and swap pair entries, preserving the dichotomy. For $x_i := x_i \wedge \neg x_i$: if α is facial via pair $j \neq i$, so is $F(\alpha)$; if via pair i, then $(x_i \wedge$

$\neg x_i, x_i \vee \neg x_i)$ evaluates at $(0,1)$ or $(1,0)$ to $(0,1)$, a facial pair. Closure under composition is immediate ($F'F(\mathrm{Face}) \subseteq F'(\mathrm{Face}) \subseteq \mathrm{Face}$), and naturality on $\square^{\mathrm{fp}}_{\mathrm{DM}}$ is Theorem 5.1(3⇒1) applied to all g. For a degeneracy $[n{+}1] \to [n]$, a point facial only via the dropped pair maps to a diagonal point; for the connection $x_1 := x_1 \wedge x_2$, the point with pairs $((1,0),(0,0),\dots)$ maps to $((0,0),(0,0),\dots)$; for the diagonal $x_1 := x_2$, the point $((0,1),(0,0),\dots)$ maps to $((0,0),(0,0),\dots)$. Violating elements exist in each case: take g with A_g containing the reached diagonal point (realizability in [10, Thm. 3.4]). $\square$

Remark 5.3 (the skeleton is not cylinder-stable)**.** The degeneracy case deserves emphasis. For the cylinder $g \times I := \sigma^* g$ along $\sigma : [n+1] \to [n]$, the two new faces are g itself; if $A_g \neq \emptyset$ then condition (3) fails (every diagonal point of Q_n is the image of a point facial only via the dropped pair), and indeed $\mathrm{r}(g \times I) \neq \mathrm{r}(g) \times I$: the minimal filler of the cylinder's boundary is not the cylinder of the minimal filler. A Reedy-style "skeleton" functor is thus unavailable in principle — consistent with, and sharpening, Proposition 3.1.

6. Thinness as structure

In cubical ω-categories with connections, Higgins [8] proved that every commutative shell has a *unique* thin filler, and Steiner [11] that the same uniqueness organizes the cubical nerves of ω-categories; in both cases "thin" is definable from the ambient composition structure (Higgins: folding into an identity; and thin structures are equivalent to connections, [8, Thm. 3.1]). Steiner remarked that for weak higher categories one expects thin fillers "to exist but not necessarily to be unique". The De Morgan strict layer is an extreme test case: there is no composition at all, fillers of a boundary form the Boolean torsor 2^D [10], and the dichotomy (Theorem 5.1) shows that no choice of representatives is stable under all reparametrizations — the obstruction being reached by degeneracies, connections and diagonals alike. What does survive is *structure*: the minimal filler (r), the defect antichain (Corollary 4.2) measuring every transport's deviation, and the median cylinder correcting it within the fiber. In slogan form: *without Eilenberg–Zilber normal forms, thinness cannot be a property; the minimal-filler-plus-median package is the minimal structure that replaces it.* We expect this package, rather than any uniqueness principle, to be the right input for algebraic weak-factorization-system treatments of the strict layer; we leave that organization to future work (§8).

7. Machine verification

All theorems above are elementary, but their statements were found, and are guarded, by exhaustive machine checks (scripts accompany the artifact; see [10] for the census infrastructure). Through $n \leq 3$: the defect formula of Theorem 4.1 was verified on all 168 elements of DM_2 and all 7,828,354 elements of DM_3 against brute-force fiber minima (zero discrepancies); the coherence dichotomy was verified on all of DM_2 for five generating substitutions (81 incoherent instances, matching $23 + 23 + 24 + 11$ counted independently), and on all of DM_3 for a diagonal, a connection, and the reversal — conditions (1), (2), (3) of Theorem 5.1 carve out identical subsets, of sizes 1,133,756, 1,133,756, and 0 respectively. At $n = 2$ the exceptional set for connections and diagonals consists exactly of the two elements whose defect is a pole of the literal cube ($g = 1$ and the four-fold conjunction $x \wedge \neg x \wedge y \wedge \neg y$); from $n = 3$ on it is large — about 14.5% of all elements for the diagonal substitution — so the dichotomy is a genuinely quantitative constraint, not an edge-case bookkeeping device.

8. Outlook

Three directions. (1) *Algebraic organization*: package (minimal filler, defect, median correction) as an algebraic weak factorization system on De Morgan cubical sets in which the boundary inclusions' right lifting structures are the canonical minimal fillers and the left class carries the Boolean fiber data; the dichotomy says any such structure will be genuinely *algebraic* over $\square_{\mathrm{DM}} \setminus \square_{\mathrm{DM}}^{\mathrm{fp}}$, never property-like — the verification of the distributivity axioms against the χ-corrections is the remaining work. (2) *The Karoubi envelope as a site*: Theorem 3.2 invites a study of presheaves on finite projective De Morgan algebras — the "involutive lattices" picture of [1] suggests comparisons both with Sattler's finite-lattice site and with the twist construction of [4], under which our literal cube is the twist of the Dedekind evaluation cube. (3) *Kleene and beyond*: every argument here uses only the literal-cube combinatorics; the Kleene case constrains but does not excise the diagonal points, and the corresponding fiber, defect and coherence theory (with [1, Thm. 12] supplying the Karoubi envelope) should interpolate between the De Morgan and Dedekind extremes, where fibers are Boolean of positive rank, respectively trivial.

Artifact. The verification scripts and the machine-checked type-theoretic counterparts of the companion results are available in the repository at `https://github.com/r-shibusawa/cubical-strict-j` (archived at DOI 10.5281/zenodo.21405961).

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